# MANGALORE REFINERY AND PETROCHEMICALS LIMITED

Reliability & Static Equipment Inspection Dept | Hydrocracker Unit-1

## API 510 STATUTORY PRESSURE VESSEL INTEGRITY REPORT

Report ID: **MRPL/PV-510/2026/V-204-INSP** | Date: **12-SEP-2026**

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|---|---|---|---|
| **Equipment Tag:** | V-204 (Amine Flash Drum) | **Design Code:** | ASME Sec VIII Div 1 / API 510 |
| **Material Spec:** | SA-516 Grade 70 (HIC Resistant) | **Operating Press / Temp:** | 18.5 bar @ 65°C |
| **Original Nominal Wall:** | 24.0 mm | **Corrosion Allowance (CA):** | 4.5 mm |
| **Retirement Thickness (t_min):** | 19.5 mm | **Service Medium:** | Rich Amine / H2S Stripped Water |

## 1. ULTRASONIC THICKNESS SURVEY & SCAN RESULTS

| Shell Zone | Nominal (mm) | Retirement t_min | Actual Measured | Loss % | Integrity Status |
|---|---|---|---|---|---|
| Top Head Crown | 24.00 mm | 19.50 mm | 22.80 mm | 5.0% | PASS |
| Middle Shell Ring 2 | 24.00 mm | 19.50 mm | 21.40 mm | 10.8% | PASS |
| Bottom Liquid-Vapor Interface | 24.00 mm | 19.50 mm | 18.20 mm | 24.1% | CRITICAL FAIL (Below t_min) |
| Boot Drain Nozzle N-3 | 16.00 mm | 12.00 mm | 14.60 mm | 8.7% | PASS |

## 2. DEGRADATION EVALUATION & STATUTORY CITATIONS

• **Mechanism:** Wet H2S sour water service accelerated localized pitting and hydrogen-induced blistering along the liquid-vapor interface line.
• **Statutory Code Violation:** Measured wall thickness at the Bottom Liquid-Vapor Interface is **18.20 mm**, falling **1.30 mm below the API 510 minimum allowable retirement thickness (19.50 mm)**. The initial 4.5 mm corrosion allowance has been completely exhausted (accumulated wall loss: 5.80 mm).
• **Risk Classification:** Level 1 Immediate Hazard - Risk of amine breakthrough and vessel loss of containment.

## 3. MANDATED CORRECTIVE ACTIONS

1. **Operational Isolation:** De-inventory vessel V-204 and route amine supply to standby surge vessel V-205 within 24 hours.
2. **Structural Repair:** Perform full internal weld overlay with 316L stainless steel cladding or insert replacement shell can per ASME PCC-2 Article 2.1.
3. **Inspection Hold Point:** Wet fluorescent magnetic particle testing (WFMT) must be performed across 100% of internal weld seams prior to hydro-testing at 27.8 bar.